Bytecode Interpreters

Concepts of Programming Languages

CAS CS 320

Outline

- Discuss **stack-based languages** and **stack machines**
- Look briefly at how to compile **variables** and **functions**

High Level

Abstract/Virtual Machines

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A **virtual machine** is a computational abstraction, like a Turing machine but *easier to implement*

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Virtual machines are often **bytecode interpreters**, "programs" are streams of bytes

One common abstraction is **stack machines**

Benefits of Abstract Machines

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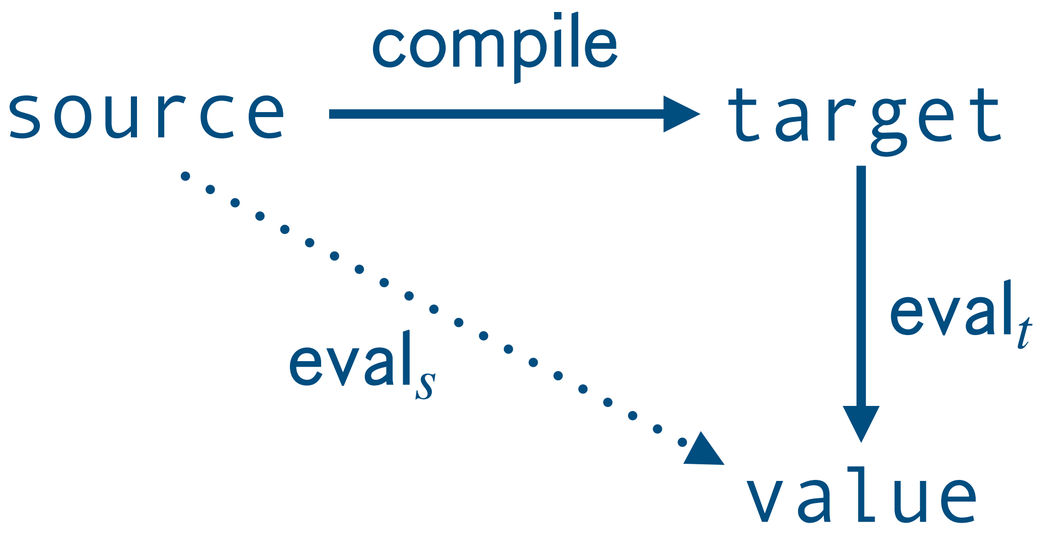
Simplicity: Things like stacks aren't too complicated

Portability: A system should be able to handle a stream of bytes

Efficiency (sort of): They can be implemented in low-level languages, and so will generally be faster than definitional interpreters (but not as fast as natively compiled code)

Compilation



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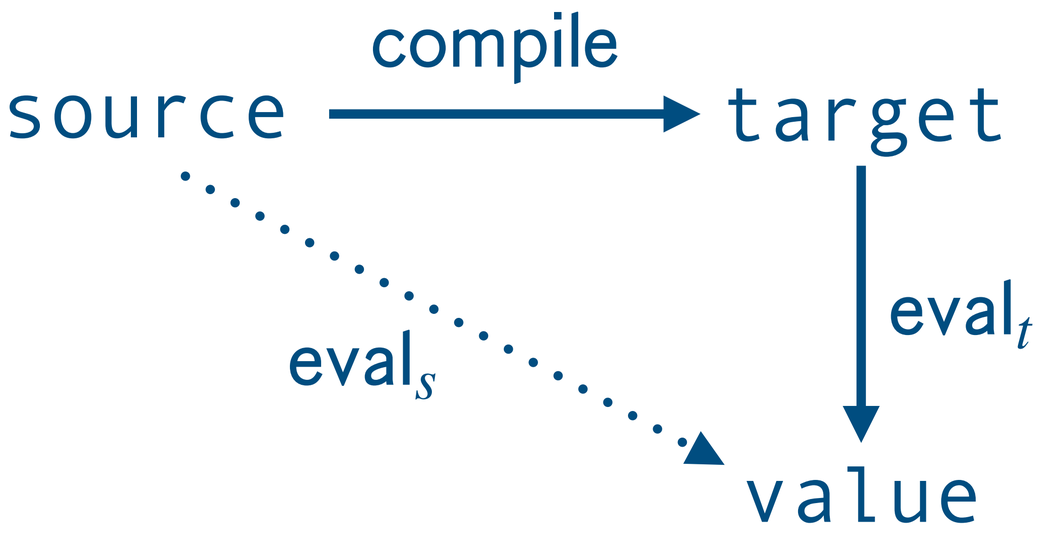
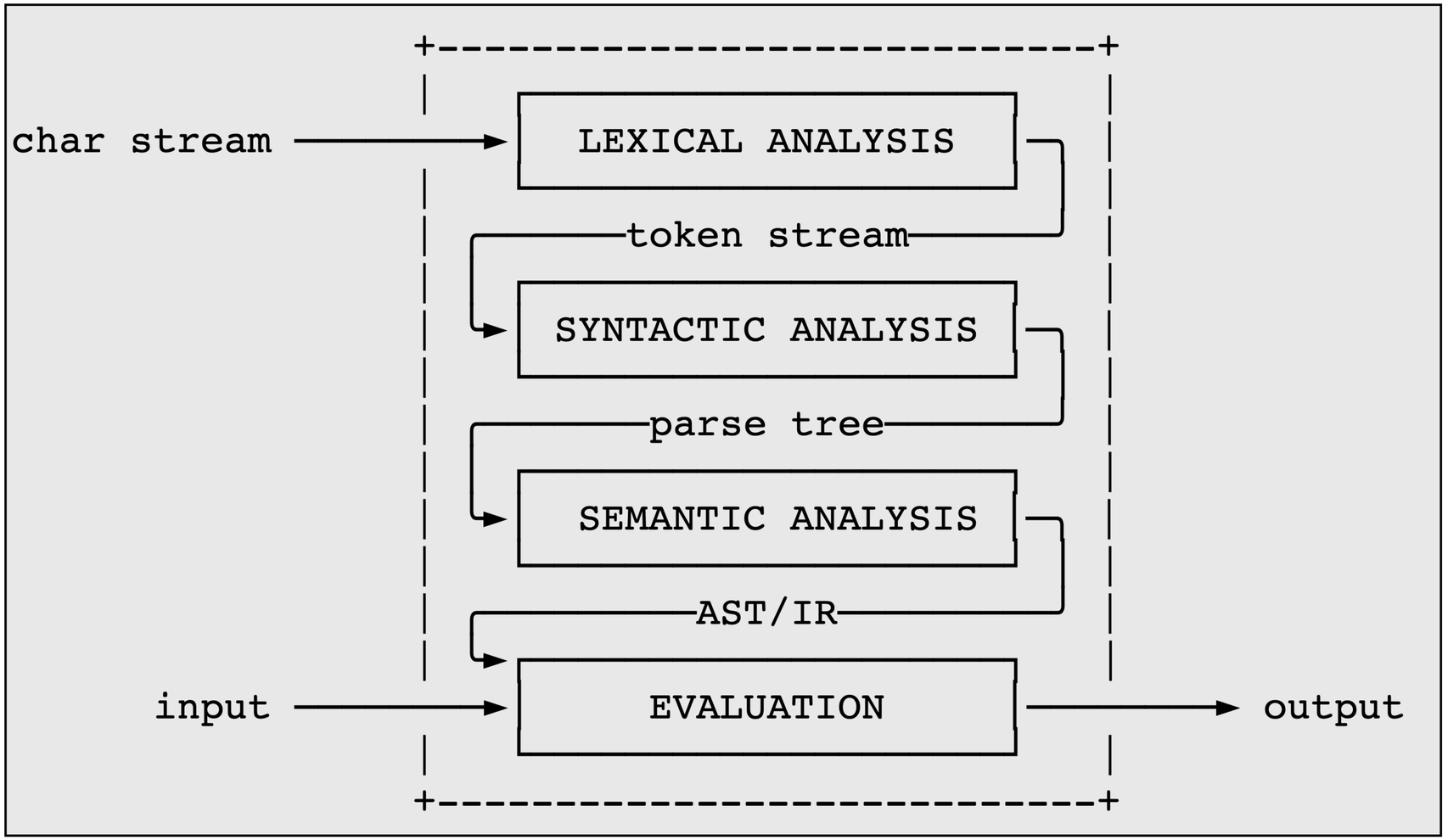
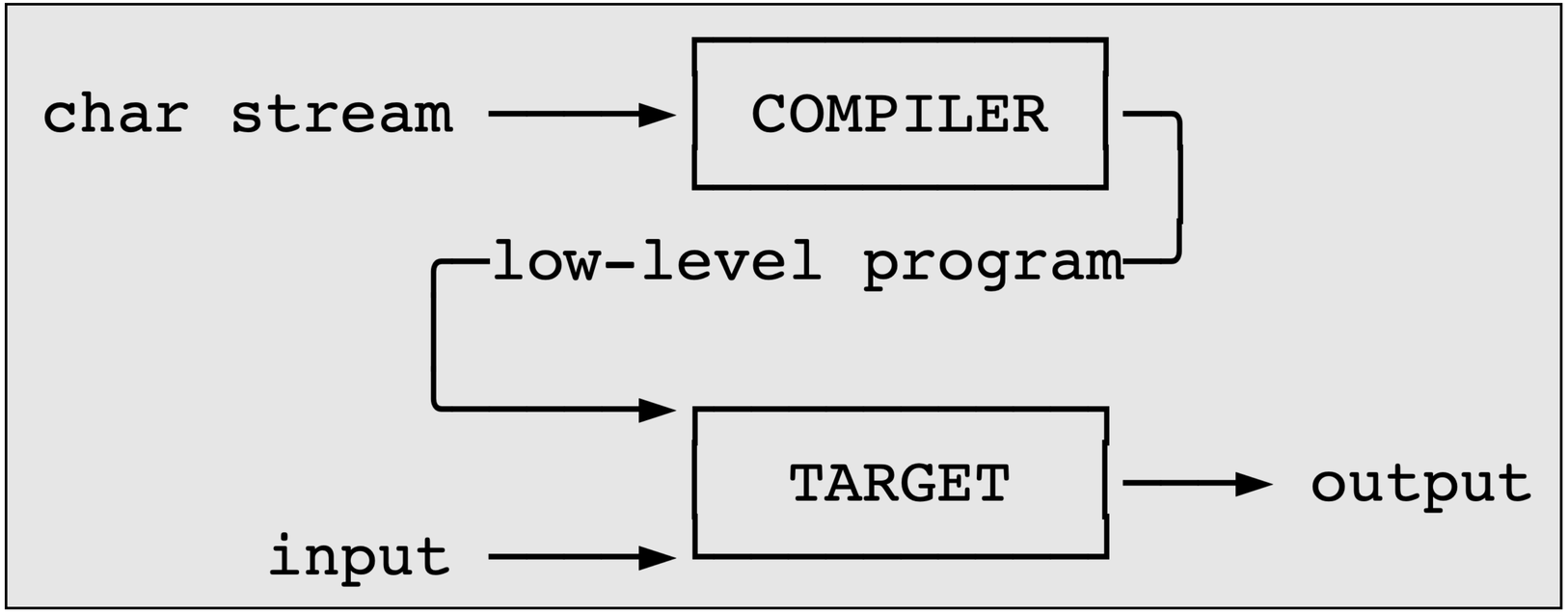


Compilation is the process of translating a program in one language to another, maintaining semantic behavior

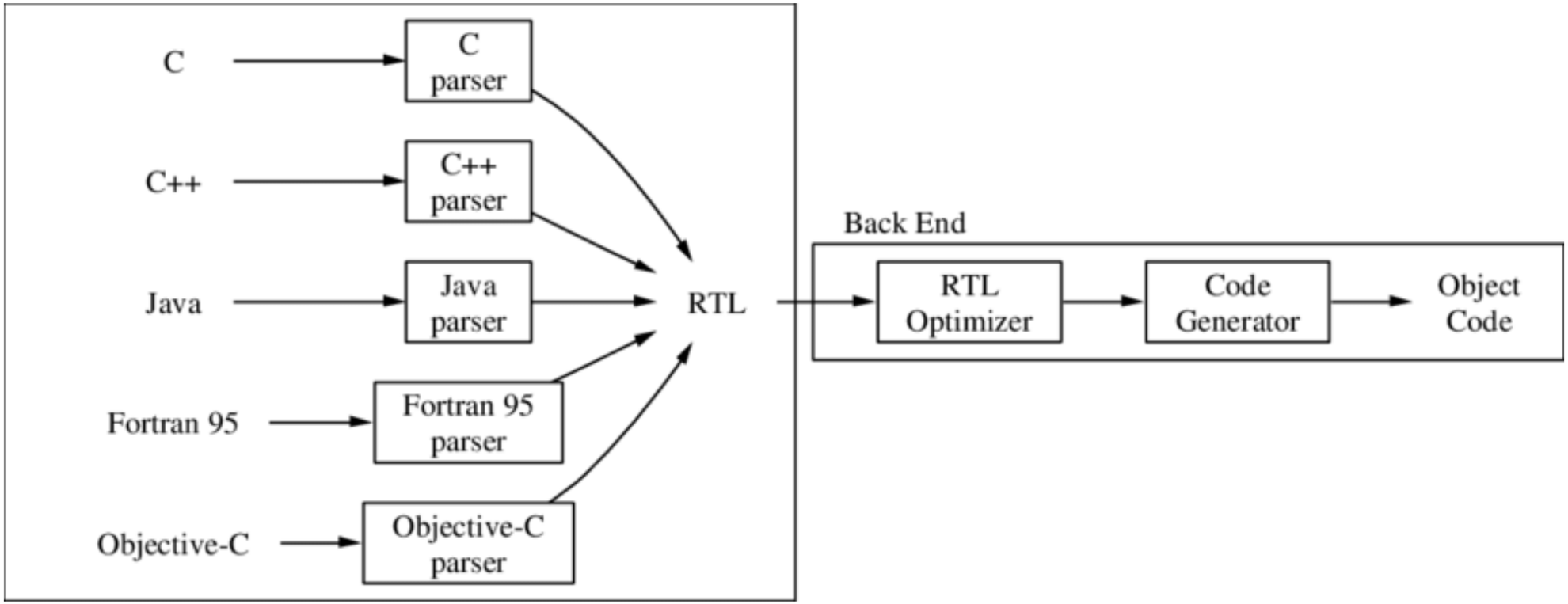
Recall: The Interpretation Pipeline



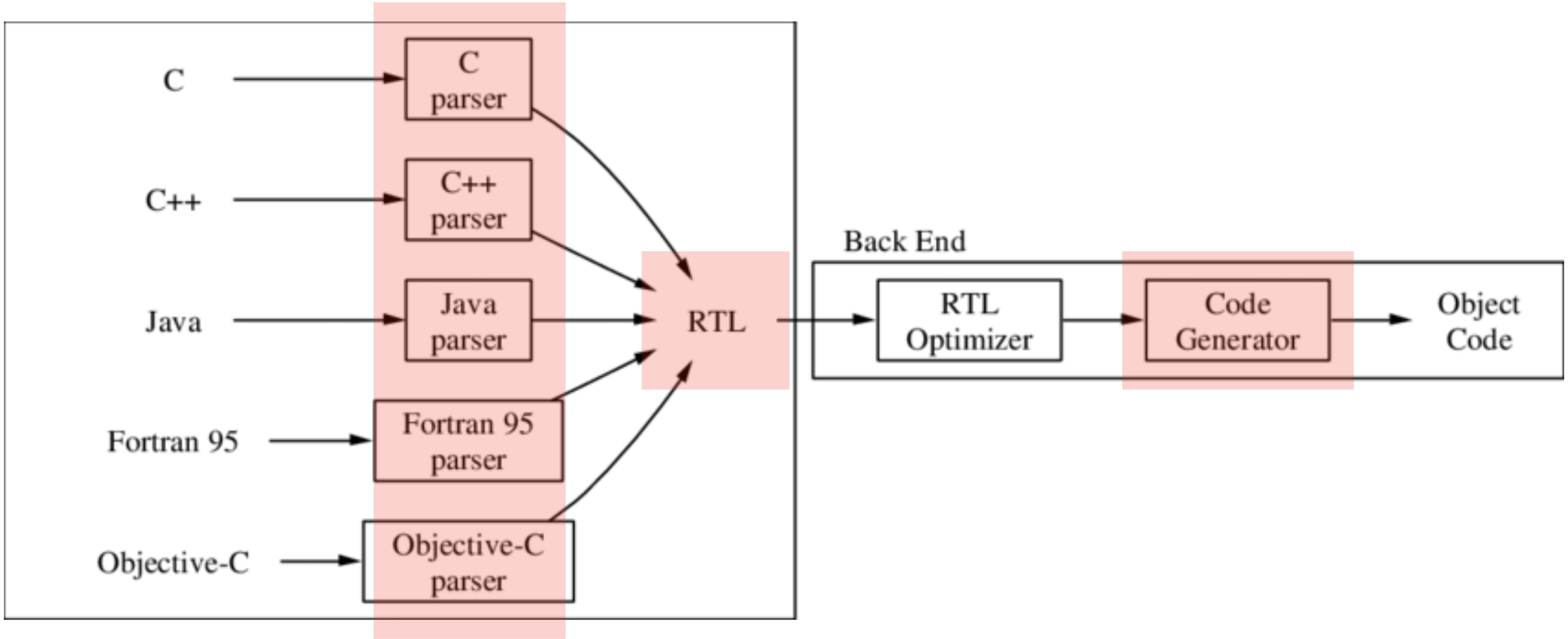
The Compiler Pipeline



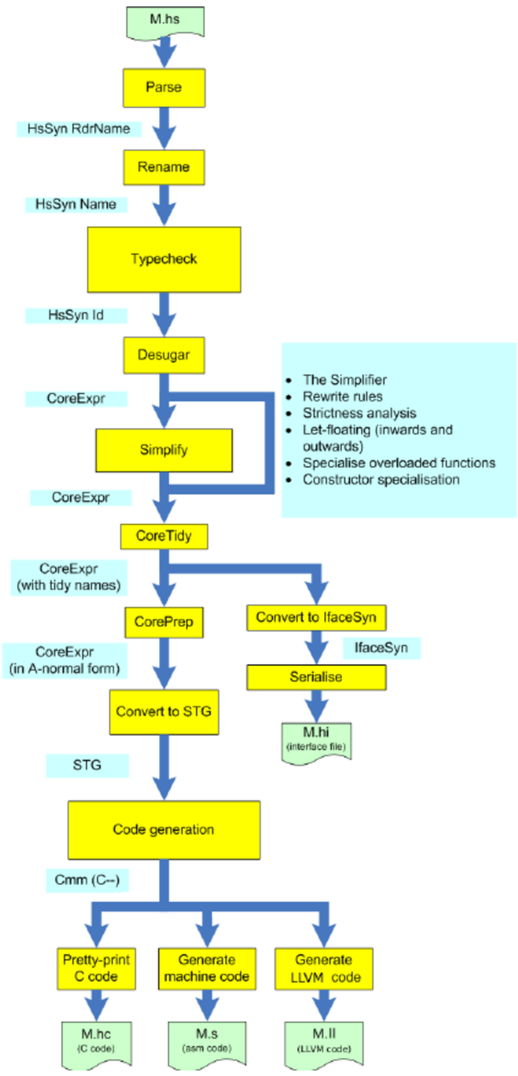
Example Pipelines: gcc



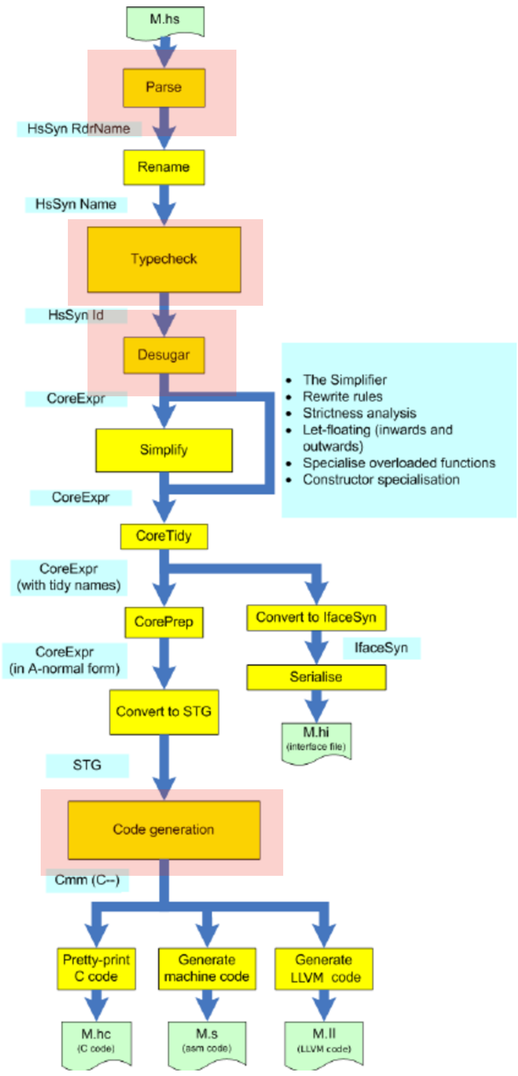
Example Pipelines: gcc



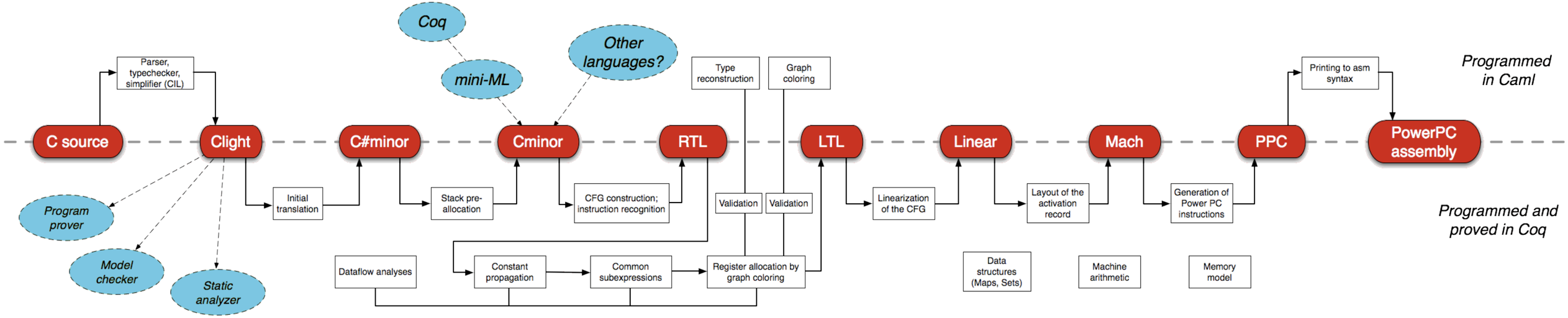
Example Pipelines: GHC (Haskell)



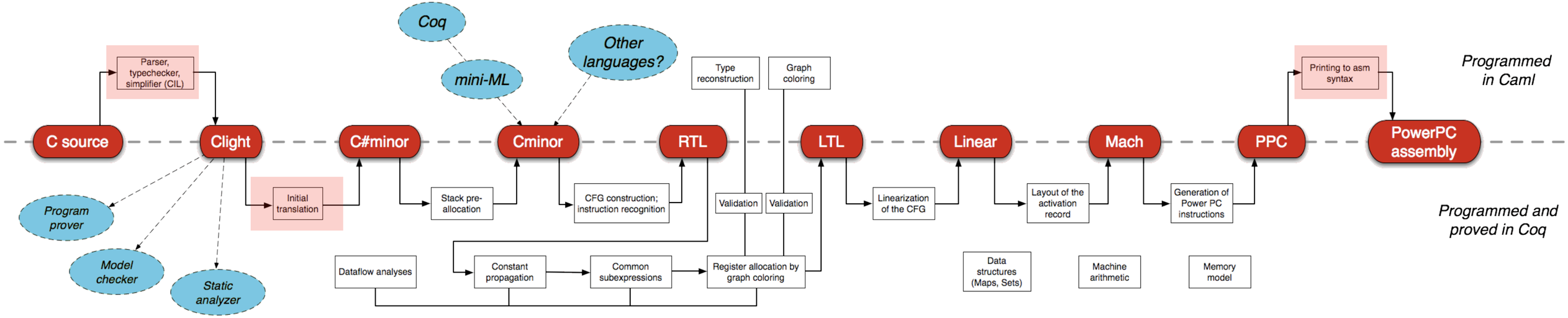
Example Pipelines: GHC (Haskell)



Example Pipelines: CompCert (C)



Example Pipelines: CompCert (C)



Stack-Based Arithmetic (Warm-up)

Stack-Based Arithmetic (Syntax)

$$
\begin{aligned}
\langle\text{prog}\rangle &::= \{\langle\text{com}\rangle\} \\
\langle\text{com}\rangle &::= \texttt{ADD} \mid \texttt{SUB} \mid \texttt{MUL} \mid \texttt{DIV} \mid \texttt{PUSH}\ \langle\text{num}\rangle \\
\langle\text{num}\rangle &::= \mathbb{Z}
\end{aligned}
$$

Stack-Based Arithmetic (Semantics)

$$\langle \mathcal{S}, P \rangle$$

A **value** is an integer ($\mathbb{Z}$)

A **configuration** is made up of a stack ($\mathcal{S}$) of values and a program (P) given by **<prog>**

Stack-Based Arithmetic (Semantics)

$$\frac{}{\langle m :: n :: \mathcal{S}, \texttt{ADD}\ P\rangle \longrightarrow \langle (m+n) :: \mathcal{S}, P\rangle}\ \texttt{(add)}$$

$$\frac{}{\langle m :: n :: \mathcal{S}, \texttt{SUB}\ P\rangle \longrightarrow \langle (m-n) :: \mathcal{S}, P\rangle}\ \texttt{(sub)}$$

$$\frac{}{\langle m :: n :: \mathcal{S}, \texttt{MUL}\ P\rangle \longrightarrow \langle (m\times n) :: \mathcal{S}, P\rangle}\ \texttt{(mul)}$$

$$\frac{n \neq 0}{\langle m :: n :: \mathcal{S}, \texttt{DIV}\ P\rangle \longrightarrow \langle (m/n) :: \mathcal{S}, P\rangle}\ \texttt{(div)}$$

$$\frac{}{\langle \mathcal{S}, \texttt{PUSH}\ n\ P\rangle \longrightarrow \langle n :: \mathcal{S}, P\rangle}\ \texttt{(push)}$$

Example (Evaluation)

$\langle \emptyset, \texttt{PUSH 2 PUSH 3 SUB PUSH 4 MUL} \rangle \longrightarrow$

$\langle 2 :: \emptyset, \texttt{PUSH 3 SUB PUSH 4 MUL} \rangle \rightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \texttt{PUSH 2 PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \texttt{PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \texttt{PUSH 2 PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \texttt{PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 3 :: 2 :: \varnothing\ ,\ \texttt{SUB PUSH 4 MUL}\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \texttt{PUSH 2 PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \texttt{PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 3 :: 2 :: \varnothing\ ,\ \texttt{SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ -1 :: \varnothing\ ,\ \texttt{PUSH 4 MUL}\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \texttt{PUSH 2 PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \texttt{PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 3 :: 2 :: \varnothing\ ,\ \texttt{SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ -1 :: \varnothing\ ,\ \texttt{PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 4 :: -1 :: \varnothing\ ,\ \texttt{MUL}\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \texttt{PUSH 2 PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

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$\langle\ -1 :: \varnothing\ ,\ \texttt{PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 4 :: -1 :: \varnothing\ ,\ \texttt{MUL}\ \rangle \longrightarrow$

$\langle\ -4 :: \varnothing\ ,\ \epsilon\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \texttt{PUSH 2 PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \texttt{PUSH 3 SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 3 :: 2 :: \varnothing\ ,\ \texttt{SUB PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ -1 :: \varnothing\ ,\ \texttt{PUSH 4 MUL}\ \rangle \longrightarrow$

$\langle\ 4 :: -1 :: \varnothing\ ,\ \texttt{MUL}\ \rangle \longrightarrow$

$\langle\ -4 :: \varnothing\ ,\ \epsilon\ \rangle \longrightarrow\ -4$

Compiling Arithmetic Expressions

$$\texttt{n} \implies \texttt{PUSH n}$$

$$e_1 \ \texttt{+} \ e_2 \implies \mathscr{C}(e_2) \ \mathscr{C}(e_1) \ \texttt{ADD}$$

$$e_1 \ \texttt{-} \ e_2 \implies \mathscr{C}(e_2) \ \mathscr{C}(e_1) \ \texttt{SUB}$$

$$e_1 \ \texttt{*} \ e_2 \implies \mathscr{C}(e_2) \ \mathscr{C}(e_1) \ \texttt{MUL}$$

$$e_1 \ \texttt{/} \ e_2 \implies \mathscr{C}(e_2) \ \mathscr{C}(e_1) \ \texttt{DIV}$$

Note the order!

Example (Compilation)

$\mathscr{C}$(**4 * (2 - 3)**) =

$\mathscr{C}$(2 - 3) $\mathscr{C}$(4) MUL =

C(2 - 3) PUSH 4 MUL =

C(3) C(2) SUB PUSH 4 MUL ...

Example (Compilation)

$\mathscr{C}$(`4 * (2 - 3)`)

$\Longrightarrow$ $\mathscr{C}$(`4`) $\mathscr{C}$(`2 - 3`) `MUL`

Example (Compilation)

$\mathscr{C}$(`4 * (2 - 3)`)

$\Longrightarrow$ $\mathscr{C}$(`4`) $\mathscr{C}$(`2 - 3`) `MUL`

$\Longrightarrow$ `PUSH 4` $\mathscr{C}$(`2 - 3`) `MUL`

Example (Compilation)

$\mathscr{C}$(`4 * (2 - 3)`)

$\Longrightarrow$ $\mathscr{C}$(`4`) $\mathscr{C}$(`2 - 3`) `MUL`

$\Longrightarrow$ `PUSH 4` $\mathscr{C}$(`2 - 3`) `MUL`

$\Longrightarrow$ `PUSH 4` $\mathscr{C}$(`2`) $\mathscr{C}$(`3`) `SUB MUL`

Example (Compilation)

$\mathscr{C}$(`4 * (2 - 3)`)

$\Longrightarrow$ $\mathscr{C}$(`4`) $\mathscr{C}$(`2 - 3`) `MUL`

$\Longrightarrow$ `PUSH 4` $\mathscr{C}$(`2 - 3`) `MUL`

$\Longrightarrow$ `PUSH 4` $\mathscr{C}$(`2`) $\mathscr{C}$(`3`) `SUB MUL`

$\Longrightarrow$ `PUSH 4 PUSH 2` $\mathscr{C}$(`3`) `SUB MUL`

Example (Compilation)

$\mathscr{C}$(`4 * (2 - 3)`)

$\Longrightarrow$ $\mathscr{C}$(`4`) $\mathscr{C}$(`2 - 3`) `MUL`

$\Longrightarrow$ `PUSH 4` $\mathscr{C}$(`2 - 3`) `MUL`

$\Longrightarrow$ `PUSH 4` $\mathscr{C}$(`2`) $\mathscr{C}$(`3`) `SUB MUL`

$\Longrightarrow$ `PUSH 4 PUSH 2` $\mathscr{C}$(`3`) `SUB MUL`

$\Longrightarrow$ `PUSH 4 PUSH 2 PUSH 2 SUB MUL`

Variables

Variables (Syntax)

$$
\begin{aligned}
\langle\text{prog}\rangle &::= \{\langle\text{com}\rangle\} \\
\langle\text{com}\rangle &::= \texttt{ADD} \mid \texttt{SUB} \mid \texttt{MUL} \mid \texttt{DIV} \mid \texttt{PUSH}\ \langle\text{num}\rangle \\
&\quad\ \mid \texttt{ASSIGN}\ \langle\text{var}\rangle \mid \texttt{LOOKUP}\ \langle\text{var}\rangle \\
\langle\text{num}\rangle &::= \mathbb{Z} \\
\langle\text{var}\rangle &::= \mathbb{I}
\end{aligned}
$$

Variables (Semantics)

$$\langle \mathcal{S}, \mathcal{E}, P \rangle$$

A **value** is an integer ($\mathbb{Z}$)

A **configuration** is made up of a stack $\mathcal{S}$ of values, an environment $\mathcal{E}$ (mapping of identifiers to values), and a program P given by **<prog>**

Variables (Semantics)

$$\frac{}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{ADD}\ P\rangle \longrightarrow \langle (m+n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ \texttt{(add)}$$

$$\frac{}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{SUB}\ P\rangle \longrightarrow \langle (m-n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ \texttt{(sub)}$$

$$\frac{}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{MUL}\ P\rangle \longrightarrow \langle (m\times n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ \texttt{(mul)}$$

$$\frac{n \neq 0}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{DIV}\ P\rangle \longrightarrow \langle (m/n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ \texttt{(div)}$$

$$\frac{}{\langle \mathcal{S}, \mathcal{E}, \texttt{PUSH}\ n\ P\rangle \longrightarrow \langle n :: \mathcal{S}, \mathcal{E}, P\rangle}\ \texttt{(push)}$$

$$\frac{}{\langle n :: \mathcal{S}, \mathcal{E}, \texttt{ASSIGN}\ x\ P\rangle \longrightarrow \langle \mathcal{S}, \mathcal{E}[x \mapsto n], P\rangle}\ \texttt{(asn)}$$

$$\frac{\mathcal{E}(x) = m}{\langle n :: \mathcal{S}, \mathcal{E}, \texttt{LOOKUP}\ x\ P\rangle \longrightarrow \langle m :: \mathcal{S}, \mathcal{E}, P\rangle}\ \texttt{(lkp)}$$

Variables (Semantics)

basically the same

$$\frac{}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{ADD}\ P\rangle \longrightarrow \langle (m+n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{add})$$

$$\frac{}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{SUB}\ P\rangle \longrightarrow \langle (m-n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{sub})$$

$$\frac{}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{MUL}\ P\rangle \longrightarrow \langle (m\times n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{mul})$$

$$\frac{n \neq 0}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{DIV}\ P\rangle \longrightarrow \langle (m/n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{div})$$

$$\frac{}{\langle \mathcal{S}, \mathcal{E}, \texttt{PUSH}\ n\ P\rangle \longrightarrow \langle n :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{push})$$

$$\frac{}{\langle n :: \mathcal{S}, \mathcal{E}, \texttt{ASSIGN}\ x\ P\rangle \longrightarrow \langle \mathcal{S}, \mathcal{E}[x \mapsto n], P\rangle}\ (\texttt{asn})$$

$$\frac{\mathcal{E}(x) = m}{\langle n :: \mathcal{S}, \mathcal{E}, \texttt{LOOKUP}\ x\ P\rangle \longrightarrow \langle m :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{lkp})$$

Variables (Semantics)

basically the same

$$\frac{}{\langle m :: n :: \mathcal{S}, \mathcal{E}, \texttt{ADD}\ P\rangle \longrightarrow \langle (m+n) :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{add})$$

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$$\frac{}{\langle \mathcal{S}, \mathcal{E}, \texttt{PUSH}\ n\ P\rangle \longrightarrow \langle n :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{push})$$

new rules

$$\frac{}{\langle n :: \mathcal{S}, \mathcal{E}, \texttt{ASSIGN}\ x\ P\rangle \longrightarrow \langle \mathcal{S}, \mathcal{E}[x \mapsto n], P\rangle}\ (\texttt{asn})$$

$$\frac{\mathcal{E}(x) = m}{\langle n :: \mathcal{S}, \mathcal{E}, \texttt{LOOKUP}\ x\ P\rangle \longrightarrow \langle m :: \mathcal{S}, \mathcal{E}, P\rangle}\ (\texttt{lkp})$$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,$ `PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD` $\rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,$ **PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD** $\rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \varnothing\ ,$ **ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD** $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\}\ ,$ **PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD** $\rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,$ **PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD** $\rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \varnothing\ ,$ **ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD** $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\}\ ,$ **PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD** $\rangle \longrightarrow$

$\langle\ 3 :: \varnothing\ ,\ \{x \mapsto 2\}\ ,$ **ASSIGN y LOOKUP x LOOKUP y ADD** $\rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 3 :: \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 3 :: \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP y ADD}\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 3 :: \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 3 :: 2 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{ADD}\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 3 :: \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 3 :: 2 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{ADD}\ \rangle \longrightarrow$

$\langle\ 5 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \epsilon\ \rangle \longrightarrow$

Example (Evaluation)

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 2 ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN x PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{PUSH 3 ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 3 :: \varnothing\ ,\ \{x \mapsto 2\}\ ,\ \texttt{ASSIGN y LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP x LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{LOOKUP y ADD}\ \rangle \longrightarrow$

$\langle\ 3 :: 2 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \texttt{ADD}\ \rangle \longrightarrow$

$\langle\ 5 :: \varnothing\ ,\ \{x \mapsto 2\ ,\ y \mapsto 3\}\ ,\ \epsilon\ \rangle \longrightarrow\ 5$

Compiling Let-Expressions (Attempt)

$$\texttt{x} \implies \texttt{LOOKUP}\ x$$

$$\texttt{let}\ x\ \texttt{=}\ e_1\ \texttt{in}\ e_2 \implies \mathscr{C}(e_1)\ \texttt{ASSIGN}\ x\ \mathscr{C}(e_2)$$

Compiling Let-Expressions (Attempt)

$$\texttt{x} \implies \texttt{LOOKUP}\ x$$

$$\texttt{let}\ x\ \texttt{=}\ e_1\ \texttt{in}\ e_2 \implies \mathscr{C}(e_1)\ \texttt{ASSIGN}\ x\ \mathscr{C}(e_2)$$

Except this isn't quite right

Example

$\mathscr{C}$(`let y = 1 in let x = let y = 2 in y in y`) $\Longrightarrow$

```
let y = 1 in
let x =
    let y = 2 in y
in y
```

Example

$\mathscr{C}$(`let y = 1 in let x = let y = 2 in y in y`) $\Longrightarrow$

$\mathscr{C}$(`1`) `ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

Example

$\mathscr{C}$(`let y = 1 in let x = let y = 2 in y in y`) $\Longrightarrow$

$\mathscr{C}$(`1`) `ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

Example

$\mathscr{C}$(`let y = 1 in let x = let y = 2 in y in y`) $\Longrightarrow$

$\mathscr{C}$(`1`) `ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let y = 2 in y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

Example

$\mathscr{C}$(`let y = 1 in let x = let y = 2 in y in y`) $\Longrightarrow$

$\mathscr{C}$(`1`) `ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let y = 2 in y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`2`) `ASSIGN y` $\mathscr{C}$(`y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

Example

$\mathscr{C}$(`let y = 1 in let x = let y = 2 in y in y`) $\Longrightarrow$

$\mathscr{C}$(`1`) `ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let y = 2 in y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`2`) `ASSIGN y` $\mathscr{C}$(`y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y PUSH 2 ASSIGN y` $\mathscr{C}$(`y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

Example

$\mathscr{C}$(`let y = 1 in let x = let y = 2 in y in y`) $\Longrightarrow$

$\mathscr{C}$(`1`) `ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let y = 2 in y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`2`) `ASSIGN y` $\mathscr{C}$(`y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y PUSH 2 ASSIGN y` $\mathscr{C}$(`y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

Example

$\mathscr{C}$(`let y = 1 in let x = let y = 2 in y in y`) $\Longrightarrow$

$\mathscr{C}$(`1`) `ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let x = let y = 2 in y in y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`let y = 2 in y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y` $\mathscr{C}$(`2`) `ASSIGN y` $\mathscr{C}$(`y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y PUSH 2 ASSIGN y` $\mathscr{C}$(`y`) `ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x` $\mathscr{C}$(`y`) $\Longrightarrow$

`PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y`

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,$ `PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 1 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,$ `PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ 1 :: \varnothing\ ,\ \varnothing\ ,$ `ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 1\}\ ,$ `PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,$ `PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ 1 :: \varnothing\ ,\ \varnothing\ ,$ `ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 1\}\ ,$ `PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 1\}\ ,$ `ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,$ **PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ 1 :: \varnothing\ ,\ \varnothing\ ,$ **ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 1\}\ ,$ **PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 1\}\ ,$ **ASSIGN y LOOKUP y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 2\}\ ,$ **LOOKUP Y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,$ `PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ 1 :: \varnothing\ ,\ \varnothing\ ,$ `ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 1\}\ ,$ `PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 1\}\ ,$ `ASSIGN y LOOKUP y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 2\}\ ,$ `LOOKUP Y ASSIGN x LOOKUP y` $\rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 2\}\ ,$ `ASSIGN x LOOKUP y` $\rangle \longrightarrow$

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,$ **PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ 1 :: \varnothing\ ,\ \varnothing\ ,$ **ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 1\}\ ,$ **PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 1\}\ ,$ **ASSIGN y LOOKUP y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 2\}\ ,$ **LOOKUP Y ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 2\}\ ,$ **ASSIGN x LOOKUP y** $\rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 2\ ,\ x \mapsto 2\}\ ,$ **LOOKUP y** $\rangle \longrightarrow$

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 1 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 1\}\ ,\ \texttt{PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 1\}\ ,\ \texttt{ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 2\}\ ,\ \texttt{LOOKUP Y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 2\}\ ,\ \texttt{ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 2\ ,\ x \mapsto 2\}\ ,\ \texttt{LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 2\ ,\ x \mapsto 2\}\ ,\ \epsilon\ \rangle \longrightarrow$

Example

$\langle\ \varnothing\ ,\ \varnothing\ ,\ \texttt{PUSH 1 ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 1 :: \varnothing\ ,\ \varnothing\ ,\ \texttt{ASSIGN y PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 1\}\ ,\ \texttt{PUSH 2 ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 1\}\ ,\ \texttt{ASSIGN y LOOKUP y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 2\}\ ,\ \texttt{LOOKUP Y ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 2\}\ ,\ \texttt{ASSIGN x LOOKUP y}\ \rangle \longrightarrow$

$\langle\ \varnothing\ ,\ \{y \mapsto 2\ ,\ x \mapsto 2\}\ ,\ \texttt{LOOKUP y}\ \rangle \longrightarrow$

$\langle\ 2 :: \varnothing\ ,\ \{y \mapsto 2\ ,\ x \mapsto 2\}\ ,\ \epsilon\ \rangle \longrightarrow\ 2$

Scoping

```
let y = 1 in
let x = let y = 2 in y in
y
```

$\nrightarrow 2$

Scoping

```
let y = 1 in
let x = let y = 2 in y in
y
```

$\nrightarrow 2$

The language we've just described is only good for compiling from languages with **dynamic scoping**

Scoping

```
let y = 1 in
let x = let y = 2 in y in
y
```

$\nrightarrow 2$

The language we've just described is only good for compiling from languages with **dynamic scoping**

We can use closures to deal with lexical scoping (and functions)

Functions

The Rough Picture

```
let k = fun x -> fun y -> x in
let a = k 2 in
a 3
```

Compilation is just a big sequence of transformations

The Rough Picture

```
(fun k ->
  let a = k 2 in
  a 3)
(fun x -> fun y -> x)
```

We can simulate let expressions with functions

The Rough Picture

```
(fun k ->
  (fun a -> a 3)
  (k 2))
(fun x -> fun y -> x)
```

and again...

The Rough Picture

```
[(fun k ->
   (fun a -> a 3)
   (k 2))
  (fun x -> fun y -> x)]
```

that was just to make the next part more convenient...
think of [expr] as as compile(expr)

The Rough Picture

```
[(fun x -> fun y -> x)]
[(fun k -> (fun a -> a 3) (k 2))]
CALL
```

We introduce as CALL command to call functions
Note the order, function/argument will go on a stack!

The Rough Picture

```
FUN ? X
  [fun y -> x]
  RETURN
[(fun k -> (fun a -> a 3) (k 2))]
CALL
```

We introduce a FUN command to define functions and a RETURN command to return from functions

The Rough Picture

```
FUN ? X
  FUN ? Y
    [x]
    RETURN
  RETURN
[(fun k -> (fun a -> a 3) (k 2))]
CALL
```

and again...

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
[(fun k -> (fun a -> a 3) (k 2))]
CALL
```

The familiar LOOKUP command...

And functions let us know how many commands they have

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN ? K
  [(fun a -> a 3) (k 2)]
  RETURN
CALL
```

and we can keep going...

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN ? K
  [k 2]
  [fun a -> a 3]
  CALL
  RETURN
CALL
```

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN ? K
  [2]
  [k]
  CALL
  [fun a -> a 3]
  CALL
  RETURN
CALL
```

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN ? K
  PUSH 2
  [k]
  CALL
  [fun a -> a 3]
  CALL
  RETURN
CALL
```

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN ? K
  PUSH 2
  LOOKUP K
  CALL
  [fun a -> a 3]
  CALL
  RETURN
CALL
```

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN ? K
  PUSH 2
  LOOKUP K
  CALL
  FUN ? A
    [a 3]
    RETURN
  CALL
  RETURN
CALL
```

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN ? K
  PUSH 2
  LOOKUP K
  CALL
  FUN ? A
    [3]
    [a]
    CALL
    RETURN
  CALL
  RETURN
CALL
```

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN ? K
  PUSH 2
  LOOKUP K
  CALL
  FUN ? A
    PUSH 3
    [a]
    CALL
    RETURN
  CALL
  RETURN
CALL
```

The Rough Picture

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN 10 K
  PUSH 2
  LOOKUP K
  CALL
  FUN 4 A
    PUSH 3
    LOOKUP A
    CALL
    RETURN
  CALL
  RETURN
CALL
```

The Rough Picture

```
let k = fun x -> fun y -> x in
let a = k 2 in
a 3
```

$\Longrightarrow$

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN 10 K
  PUSH 2
  LOOKUP K
  CALL
  FUN 4 A
    PUSH 3
    LOOKUP A
    CALL
    RETURN
  CALL
  RETURN
CALL
```

Compilation is just a big sequence of transformations

The Rough Picture

```
let k = fun x -> fun y -> x in
let a = k 2 in
a 3
```

$\Longrightarrow$

```
7 4
7 2
10 1
9
9
7 10
0 2
10 0
8
7 4
0 3
10 0
8
9
8
9
8
```

Byte-code interpretation additionally maps each command to a byte value

Syntax

```
<prog> ::= { <com> }
<com>  ::= PUSH <int>
         | ADD | SUB | MUL | DIV
         | FUN <ident> <int> | CALL | RETURN
         | LOOKUP
```

Semantics (Configurations)

$$\langle \mathcal{S}, \mathcal{E}, P \rangle$$

A **value** is an integer ($\mathbb{Z}$) or a closure ($\mathbb{C}$) of the form $(\mathcal{E}, x, P)$

A **configuration** is made up of a stack $\mathcal{S}$ of values, an environment $\mathcal{E}$ (mapping of identifiers to values) and a program P given by **<prog>**

Semantics (Functions)

$$\frac{}{\langle \mathcal{S}, \mathcal{E}, \mathsf{FUN}\ x\ n\ P \rangle \longrightarrow \langle (\mathcal{E}, x, P[1..n]) :: \mathcal{S}, \cancel{\mathcal{E}[x \mapsto v]}, P[n+1..] \rangle} \text{(fun)}$$

$\mathcal{E}$

Function definitions carry a **parameter name** and an **offset,** which we use to construct the closure

Semantics (Continuation Passing)

$$\frac{}{\langle (\mathscr{E}', x, P') :: v :: \mathscr{S}, \mathscr{E}, \texttt{CALL}\ P \rangle \longrightarrow \langle (\mathscr{E}, _, P) :: \mathscr{S}, \mathscr{E}'[x \mapsto v], P' \rangle}\ \texttt{(call)}$$

$$\frac{}{\langle v :: (\mathscr{E}', _, P') :: \mathscr{S}, \mathscr{E}, \texttt{RETURN}\ P \rangle \longrightarrow \langle v :: \mathscr{S}, \mathscr{E}', P' \rangle}\ \texttt{(ret)}$$

One challenge: when we call a function, where to we "return" to?

Answer: We put the information on the stack itself in a closure!

Example

Stack:

Env:

```
FUN 4 X
  FUN 2 Y
    LOOKUP X
    RETURN
  RETURN
FUN 10 K
  PUSH 2
  LOOKUP K
  CALL
  FUN 4 A
    PUSH 3
    LOOKUP A
    CALL
    RETURN
  CALL
  RETURN
CALL
```

Example

Stack:

```
Ø
x
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

Env:

```
FUN 10 K
  PUSH 2
  LOOKUP K
  CALL
  FUN 4 A
    PUSH 3
    LOOKUP A
    CALL
    RETURN
  CALL
  RETURN
CALL
```

Example

Stack:

```
Ø
X
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

```
Ø
K
PUSH 2
LOOKUP K
CALL
FUN 4 A
  PUSH 3
  LOOKUP A
  CALL
  RETURN
CALL
RETURN
```

Env:

CALL

Example

Stack:

$\emptyset$
–
ϵ

Env:

K $\mapsto$

$\emptyset$
x
FUN 2 Y
LOOKUP X
RETURN
RETURN

```
PUSH 2
LOOKUP K
CALL
FUN 4 A
  PUSH 3
  LOOKUP A
  CALL
  RETURN
CALL
RETURN
```

Example

Stack:

$\emptyset$
–
ϵ

2

Env:

K $\mapsto$

$\emptyset$
x
FUN 2 Y
  LOOKUP X
  RETURN
RETURN

```
LOOKUP K
CALL
FUN 4 A
  PUSH 3
  LOOKUP A
  CALL
  RETURN
CALL
RETURN
```

Example

Stack:

```
Ø
–
ϵ
```

```
2
```

```
Ø
x
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

Env:

K ↦

```
Ø
x
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

```
CALL
FUN 4 A
  PUSH 3
  LOOKUP A
  CALL
  RETURN
CALL
RETURN
```

Example

Stack:

$\emptyset$

–

ϵ

```
K ↦ {...}

–
FUN 4 A
  PUSH 3
  LOOKUP A
  CALL
  RETURN
CALL
RETURN
```

Env:

$X \mapsto \boxed{2}$

```
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

Example

Stack:

$\emptyset$

–

ϵ

```
K ↦ {...}

–
FUN 4 A
  PUSH 3
  LOOKUP A
  CALL
  RETURN
CALL
RETURN
```

```
X ↦ 2
Y
LOOKUP X
RETURN
```

Env:

$X \mapsto \boxed{2}$

RETURN

Example

Stack:

```
∅
–
ϵ
```

```
X ↦ 2
Y
LOOKUP X
RETURN
```

Env:

K ↦

```
∅
X
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

```
FUN 4 A
  PUSH 3
  LOOKUP A
  CALL
  RETURN
CALL
RETURN
```

Example

Stack:

```
∅
–
ϵ
```

```
X ↦ 2
Y
LOOKUP X
RETURN
```

```
K ↦ {...}
A
PUSH 3
LOOKUP A
CALL
RETURN
```

Env:

K ↦

```
∅
X
FUN 2 Y
   LOOKUP X
   RETURN
RETURN
```

```
CALL
RETURN
```

Example

Stack:

```
∅
_
ϵ
```

```
K ↦ {...}
_
RETURN
```

Env:

K ↦

```
∅
X
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

A ↦

```
X ↦ 2
Y
LOOKUP X
RETURN
```

```
PUSH 3
LOOKUP A
CALL
RETURN
```

Example

Stack:

```
Ø
_
ϵ
```

```
K ↦ {...}
_
RETURN
```

```
3
```

Env:

K ↦

```
Ø
X
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

A ↦

```
X ↦ 2
Y
LOOKUP X
RETURN
```

```
LOOKUP A
CALL
RETURN
```

Example

Stack:

```
Ø
_
ϵ
```

```
K ↦ {...}
_
RETURN
```

```
3
```

```
X ↦ 2
Y
LOOKUP X
RETURN
```

Env:

K ↦

```
Ø
X
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

A ↦

```
X ↦ 2
Y
LOOKUP X
RETURN
```

```
CALL
RETURN
```

Example

Stack:

$\emptyset$

–

ϵ

K ↦ {...}

–

RETURN

K ↦ {...}
A ↦ {...}

–

RETURN

Env:

X ↦ 2

Y ↦ 3

LOOKUP X
RETURN

Example

Stack:

$\emptyset$

–

ϵ

K ↦ {...}

–

RETURN

K ↦ {...}

A ↦ {...}

–

RETURN

2

Env:

X ↦ 2

Y ↦ 3

RETURN

Example

Stack:

```
Ø
_
ϵ
```

```
K ↦ {...}
_
RETURN
```

```
2
```

Env:

K ↦

```
Ø
X
FUN 2 Y
  LOOKUP X
  RETURN
RETURN
```

A ↦

```
X ↦ 2
Y
LOOKUP X
RETURN
```

```
RETURN
```

Example

Stack:

$\emptyset$
–
ϵ

2

Env:

K $\mapsto$

$\emptyset$
x
FUN 2 Y
  LOOKUP X
  RETURN
RETURN

RETURN

Example

<u>Stack:</u>

$\boxed{2}$

<u>Env:</u>

ϵ

demo

(show-and-tell)